

MEDIATEK

MT8676 Hypervisor I2C User Manual

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Version History

Version	Date	Author	Description
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1.1	2024-12-27	RyanW Wang	Added Section 1.3.3 Interrupt Configuration Instruction and 1.3.4 Virtual I2C Configuration
1.2	2025-03-17	Shunchang Wang	Added Section 1.3.5 Node Configuration and Section 1.3.6 Configuration for Special Application Requirements

Table of Contents

Version History 2

Table of Contents 3

List of Figures 3

List of Tables 3

1 I2C 4

1.1 Overview 4

1.1.1 Brief Introduction 4

1.1.2 Abbreviations 4

1.2 Architecture/Process Overview 4

1.2.1 I2C Introduction 5

1.2.2 MT8676 I2C Features 5

1.2.3 I2C Transmission Format 5

1.3 Configuration/Customization Guideline 6

1.3.1 DTS Configuration 6

1.3.2 Frequency 6

1.3.3 Interrupt Configuration Instruction 7

1.3.4 Virtual I2C Configuration 7

1.3.5 Node Configuration 8

1.3.6 Configuration for Special Application Requirements 8

1.4 Frequently Asked Questions/Troubleshooting 9

1.4.1 I2C Issue Debugging Methods 9

1.4.2 GPIO Mode Check 9

1.4.3 Waveform Measurement 9

1.4.4 How to Print I2C Register Information 9

1.4.5 MediaTek Support Seeking 10

Exhibit 1 Terms and Conditions 11

List of Figures

Figure 1-1. Pin connection between I2C master and I2C slave 4

Figure 1-2. I2C transmission format 5

List of Tables

Table 1-1. Abbreviations 4

1 I2C

1.1 Overview

1.1.1 Brief Introduction

This chapter introduces the hardware, software and functions of the MT8676 I2C controller.

1.1.2 Abbreviations

Table 1-1. Abbreviations

Abbreviation	Explanation
DMA	Direct Memory Access
FIFO	First Input First Output
FM	Fast Mode
FM+	Fast Mode Plus
GPIO	General Purpose Inputs-Outputs
HS mode	High Speed Mode
I/O	Input/Output
I2C	Inter-IC
SCL	Serial Clock Line
SDA	Serial Data Line
SM	Standard Mode

1.2 Architecture/Process Overview

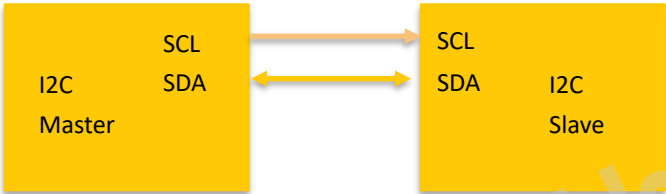


Figure 1-1. Pin connection between I2C master and I2C slave

1.2.1 I2C Introduction

The I2C controller is a bi-directional, two-wire serial interface that utilizes Serial Clock Line (SCL) and Serial Data Line (SDA) signals. These signals are capable of being driven by either the master or the slave in I2C. This generic controller supports the master role and conforms to the I2C specification.

1.2.2 MT8676 I2C Features

- Start, repeated start and stop conditions generation
- Bus detection
- Acknowledge bit generation and detection
- 7-bit/10-bit addressing
- Clock stretching
- Active drive/wired-and I/O configuration
- I2C has four operating speeds:
 - I2C Standard mode (SM) with speed up to 100 kbit/s
 - I2C Fast mode (FM) with speed up to 400 kbit/s
 - I2C Fast mode Plus (FM+) with speed up to 1 Mbit/s
 - I2C High Speed (HS) mode with speed up to 3.4 Mbit/s
- Adjustable clock speed for SM/FM/FM+/HS mode operation
- FIFO mode and DMA mode
- Multiple transfer formats:
 - Multi-write per transfer
 - Multi-read per transfer
 - Multi-transfer per transaction
 - Combined format transfer with length change capability
 - Combined format transfer with direction change capability
 - Multi-transfer with repeated start condition

1.2.3 I2C Transmission Format

Figure 1-2 illustrates the basic transfer format utilized by the SM/FM/FM+ of I2C. Initially, the master sends a start condition. Following this, the master sends the 7-bit static address of the I2C slave device with which it intends to communicate. Once the slave responds to the addressing, the master sends/receives the data. Upon completion of the data transfer, the master sends a stop condition and the bus returns to the free state.

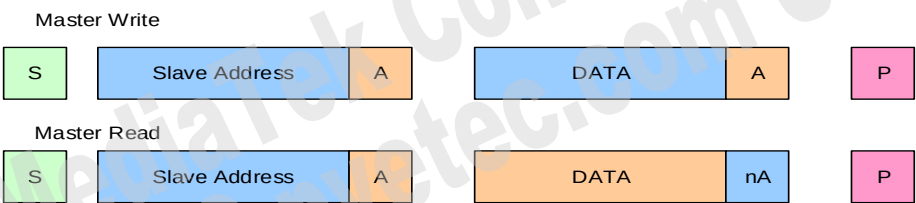


Figure 1-2. I2C transmission format

1.3 Configuration/Customization Guideline

1.3.1 DTS Configuration

Path: src/kernel/linux/v-xx/arch/arm64/boot/dts/mediatek/xxxx.dts

- Add I2C pin to set I2C pinmux

```
&pio {
    i2c2_pins: i2c2-default {
        pins-bus {
            pinmux = <PINMUX_GPIO188__FUNC_SCL2>,
                <PINMUX_GPIO189__FUNC_SDA2>;
            bias-pull-up = <MTK_PULL_SET_RSEL_111>;
        };
    };
};
```

- Add slave device node

```
&i2c2 {
    status = "okay";
    pinctrl-names = "default";
    pinctrl-0 = <&i2c2_pins>;

    nxp_eusb2_repeater_p2: nxp-eusb2-repeater-p2@4f {
        compatible = "mtk,nxp-eusb2-repeater";
        pinctrl-names = "enable", "disable";
        pinctrl-0 = <&ptn3222_enable_p2>;
        pinctrl-1 = <&ptn3222_disable_p2>;
        reg = <0x4f>;
        #phy-cells = <0>;
        status = "okay";
    };
};
```

1.3.2 Frequency

I2C controller supports data rates of 100 Kbps, 400 Kbps, and 1 Mbps.

Add **clock-frequency** = <xxxxxx> to set I2C frequency.

```
&i2c2 {
    status = "okay";
    pinctrl-names = "default";
    pinctrl-0 = <&i2c2_pins>;
    clock-frequency = <400000>;
};
```

1.3.3 Interrupt Configuration Instruction

When the MT8676 Hypervisor I2C operates in interrupt passthrough mode, the interrupt is by default directed to the Yocto side. If the interrupt is configured to the Android side, access to this interrupt on the Yocto side becomes unavailable. If you wish to modify the I2C interrupt in this manner, please seek assistance from MediaTek.

1.3.4 Virtual I2C Configuration

Path: src/kernel/linux/v-xx/arch/arm64/boot/dts/mediatek/xxxx.dts

- Virtual I2C nodes

```
i2c0_virtio:i2c0-virtio{
    compatible = "mediatek,virtio-i2c";
    id = <0>;
    status = "disabled";
};

i2c1_virtio:i2c1-virtio{
    compatible = "mediatek,virtio-i2c";
    id = <0>;
    status = "disabled";
};

i2c2_virtio:i2c2-virtio{
    compatible = "mediatek,virtio-i2c";
    id = <0>;
    status = "disabled";
};
```

- Add slave device on virtual I2C bus

```
&i2c2_virtio {
    status = "okay";

    nxp_eusb2_repeater_p2: nxp-eusb2-repeater-p2@4f {
        compatible = "mtk,nxp-eusb2-repeater";
        pinctrl-names = "enable", "disable";
        pinctrl-0 = <&ptn3222_enable_p2>;
        pinctrl-1 = <&ptn3222_disable_p2>;
        reg = <0x4f>;
        #phy-cells = <0>;
        status = "okay";
    };
};
```

1.3.5 Node Configuration

The master node should be set to “disabled” by default.

```
i2cX: i2c@xxxx {
    compatible = "mediatek,mt6897-i2c";
    ....
    status = "disabled";
};
```

The status of the corresponding device node should be set to “okay” according to which project to use.

```
&i2cX {
    status = "okay";
};
```

1.3.6 Configuration for Special Application Requirements

If there is a power consumption requirement and you need to configure I2C to a low level during suspend, the following method is required:

```
&i2cX {
    status = "okay";
    pinctrl-names = "default", "sleep";
    pinctrl-0 = <&i2cX_default>;
    pinctrl-1 = <&i2cX_sleep>;
};

&pio {
    i2cX_default: i2cX-pins-default {
        pins-bus {
            pinmux = <PINMUX_GPIOxxx_FUNC_SCLX>,
                <PINMUX_GPIOxxx_FUNC_SDAX>;
            bias-pull-up = <MTK_PULL_SET_RSEL_111>;
        };
    };

    i2c3_sleep: i2c3-pins-sleep {
        pins-bus {
            pinmux = <PINMUX_GPIOxxx_FUNC_SCLX>,
                <PINMUX_GPIOxxx_FUNC_SDAX>;
            bias-pull-down = <MTK_PULL_SET_RSEL_111>;
        };
    };
};
```

Note: This method is only for the configuration method using internal pull-up. External pull-up needs to achieve this function by controlling the power supply of the external pull-up.

1.4 Frequently Asked Questions/Troubleshooting

1.4.1 I2C Issue Debugging Methods

Most I2C problems can be debugged by trying the following methods:

- Check the slave device initialization and power supply
- Check whether the DTS configuration and GPIO mode attributes are correct
- Check whether the I2C register information is correct (you can provide log to RD)

1.4.2 GPIO Mode Check

To view the status of pins in the Kernel, enter the following commands:

```
# cd /sys
# find -name mt_gpio
# cat mt_gpio //
```

For example:

```
# cat /sys/devices/platform/soc/1000b000.pinctrl/mt_gpio
PIN: [MODE] [DIR] [DOUT] [DIN] [PULL_EN] [PULL_SEL] [IES] [SMT] [DRIVE] ( [R1] [R0] )
0: 0 0 0 0 1 0 1 0 0
1: 0 0 0 0 1 0 1 0 0
2: 0 1 1 1 1 0 1 0 0
3: 6 0 0 0 1 0 1 0 0
```

1.4.3 Waveform Measurement

Use an oscilloscope to measure if the I2C waveforms meet expectations.

1.4.4 How to Print I2C Register Information

Add the following content in `driver/i2c/busses/i2c-mt65xx.c`

```
static void i2c_dump_register(struct mtk_i2c *i2c)
{
    dev_dbg(i2c->dev, "SLAVE_ADDR: 0x%x, INTR_MASK: 0x%x\n",
            mtk_i2c_readw(i2c, OFFSET_SLAVE_ADDR),
            mtk_i2c_readw(i2c, OFFSET_INTR_MASK));
    dev_dbg(i2c->dev, "INTR_STAT: 0x%x, CONTROL: 0x%x\n",
            mtk_i2c_readw(i2c, OFFSET_INTR_STAT),
            mtk_i2c_readw(i2c, OFFSET_CONTROL));
}
```

Modify `dev_dbg` as **`dev_err`**.

1.4.5 MediaTek Support Seeking

If you are still unable to resolve your issue after the troubleshooting steps mentioned earlier, and when seeking help from MediaTek, please also provide the relevant logs containing I2C register information, waveform charts, DTS, and the output from `cat mt_gpio` as mentioned previously.

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